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Test Name:	**Mock Test**
Taken On:	4 Aug 2025 19:25:23 IST
Time Taken:	7 min 17 sec/ 22 min
Invited by:	Ankush
Invited on:	4 Aug 2025 19:25:16 IST
Skills Score:	
Tags Score:	Algorithms 105/105 Core CS 105/105 Easy 105/105 Problem Solving 105/105 Strings 105/105 problem-solving 105/105

100%

105/105

scored in **Mock Test** in 7 min 17 sec on 4 Aug 2025 19:25:23 IST

Recruiter/Team Comments:

No Comments.

	Question Description	Time Taken	Score	Status
Q1	Palindrome Index > **Coding**	7 min 6 sec	105/ 105	✓

QUESTION 1

Correct Answer

Score 105

Palindrome Index > Coding

Strings Algorithms Easy problem-solving Core CS Problem Solving

QUESTION DESCRIPTION

Given a string of lowercase letters in the range ascii[a-z], determine the index of a character that can be removed to make the string a palindrome. There may be more than one solution, but any will do. If the word is already a palindrome or there is no solution, return *-1*. Otherwise, return the index of a character to remove.

Example

$s = \text{"bcbc"}$

Either remove *'b'* at index 0 or *'c'* at index 3.

Function Description

Complete the *palindromeIndex* function in the editor below.

palindromeIndex has the following parameter(s):

- *string s:* a string to analyze

Returns

- *int:* the index of the character to remove or -1

Input Format

The first line contains an integer q, the number of queries.
Each of the next q lines contains a query string s.

Constraints

- $1 \leq q \leq 20$
- $1 \leq \text{length of } s \leq 10^5 + 5$
- All characters are in the range ascii[a-z].

Sample Input

```
STDIN   Function
-----   --------
3       q = 3
aaab    s = 'aaab' (first query)
baa     s = 'baa'  (second query)
aaa     s = 'aaa'  (third query)
```

Sample Output

```
3
0
-1
```

Explanation

Query 1: "aaab"
Removing *'b'* at index 3 results in a palindrome, so return 3.

Query 2: "baa"
Removing *'b'* at index 0 results in a palindrome, so return 0.

Query 3: "aaa"
This string is already a palindrome, so return -1. Removing any one of the characters would result in a palindrome, but this test comes first.

Note: The custom checker logic for this challenge is available here.

CANDIDATE ANSWER

Language used: **C++14**

```


// palindrom test

#include <string>

bool is_palin(string s)
{
    auto sc = s;
    std::reverse(sc.begin(), sc.end());
    return (sc==s);
}

bool is_palin(string s, int id)
{
```

```cpp
    int n = s.length();
    if (id<0 || id>=n) return false;

    // (1, .., n-1) case
    if (id==0) return is_palin(s.substr(1, n-1));
    // (0, .., n-2) case
    else if (id==n-1) return is_palin(s.substr(0, n-1));
    else {
        auto sr = s.substr(0, id) + s.substr(id+1);
        return is_palin(sr);
    }
    return false;
}

int pal_ind(string s)
{
    if (is_palin(s)) return -1;

    auto sr = s;
    std::reverse(sr.begin(), sr.end());

    int n = s.length();
    for(int i=0;i<n;i++) {
        if (s[i] != sr[i]) {
            if (is_palin(s, i)) return i;
            else if (is_palin(s, n-1-i)) return n-1-i;
            else return -1;
        }
    }
    return -1;
}

/*
 * Complete the 'palindromeIndex' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts STRING s as parameter.
 */

int palindromeIndex(string s) {
    return pal_ind(s);
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
Testcase 1	Easy	Sample case	Success	0	0.0109 sec	8.38 KB
Testcase 2	Medium	Hidden case	Success	5	0.0087 sec	8.63 KB
Testcase 3	Medium	Hidden case	Success	5	0.0096 sec	8.63 KB
Testcase 4	Medium	Hidden case	Success	5	0.0095 sec	8.63 KB
Testcase 5	Medium	Hidden case	Success	5	0.0083 sec	8.75 KB
Testcase 6	Medium	Hidden case	Success	5	0.0155 sec	8.96 KB
Testcase 7	Medium	Hidden case	Success	5	0.0169 sec	8.77 KB
Testcase 8	Medium	Hidden case	Success	5	0.0213 sec	8.72 KB
Testcase 9	Hard	Hidden case	Success	10	0.0152 sec	9.09 KB
Testcase 10	Hard	Hidden case	Success	10	0.0139 sec	8.96 KB

Testcase 11	Hard	Hidden case	Success	10	0.0186 sec	8.99 KB
Testcase 12	Hard	Hidden case	Success	10	0.012 sec	8.25 KB
Testcase 13	Hard	Hidden case	Success	10	0.0154 sec	8.85 KB
Testcase 14	Hard	Hidden case	Success	10	0.0208 sec	8.88 KB
Testcase 15	Hard	Hidden case	Success	10	0.0163 sec	8.98 KB

No Comments

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